

Ch4_The Financial Risk & Disaster Planning

Overview of FI

What is Financial Institutions (FI)?

A **company** engaged in the business of dealing with **financial** and **monetary transactions**.

- Deposit – 存款
- Loan – 借贷

Some depositors need their money at any given moment, most do not. So banks can use deposits to make long-term loans.

- Investment
- Currency exchange

They act as **intermediaries** between those who have surplus funds (savers/investors) and those who need funds(borrowers/spenders)

There are four main types of financial services: **commercial banks, credit unions, insurance companies, and investment firms**. Commercial banks are the most common type of financial institution. They offer a full range of services, including checking and savings accounts, loans, mortgages, and credit cards.

Fundamental purposed of FI

- Facilitate the **Efficient Flow** of Money in an economy.
- Reduce Transaction Costs and Manage Risk. 没有银行，手提面交
- Promote Economic Growth by channeling savings into productive investments
通过将储蓄引导到生产性投资中来促进经济增长

金融系统的一个核心作用是把人们存起来的钱（储蓄）转化为能创造价值的项目（投资），从而推动整个经济的发展。

比如：

- 储户把钱存进银行；
- 银行把这笔钱贷给企业去建工厂、开发新产品；

Core Functions of Financial Institutions

1.1. Maturity Transformation – 期限转换

Borrowing short–term and **lending long**–term .This requires managing the associated risk. 金融机构从客户那里**借入短期**资金（如可随时提取的存款），再将**这些资金用于长期贷款**（如30年期房贷）。

- 客户存款（尤其是活期存款）利率低，银行的融资成本也低。
- 如果银行用这些低成本资金去发放长期贷款（如房贷），可以赚取更高的利差（spread）
- 如果银行借的是长期资金，却只发放短期贷款，资金利用效率低，利差也小。

1.2. Risk Transformation

Pooling and **diversifying risk** across many customers (e.g., an insurance company pools premiums to pay for the losses of a few).

1.3. Reduction of Transaction Costs

Lowering the cost of **moving money** between parties through **economies of scale and expertise**.

- **Economies of Scale** – 当企业或机构的生产规模扩大时，单位成本下降，效率提升，这种现象就叫规模效益。As a company or institution **expands** its **production scale**, the **average cost per unit** decreases.
- **Specialization** – 将任务分工给最擅长的人或系统，从而提高效率和质量

1.4. Payment processing

Providing the systems and services (e.g., checks, wire transfers, debit/credit cards) to facilitate the settlement of transactions.

提供各种支付工具和系统

- **Checks** 支票
- **Wire transfer** 电汇 – An electronic way to send money from one bank account to another
- **Debit card** 借记卡 – When you pay, money is **immediately deducted** from your account.
- **Credit card** – You borrow money from the bank to make purchases, then pay it back later.

1.5. Information processing

Collecting and analyzing information about **borrowers** to assess评估**creditworthiness**, reducing the problem of asymmetric information.

- Creditworthiness: Measures how reliable you are in repaying your debts

- Character, capacity, capital, collateral and conditions are the 5 C's of credit.

- Characters – Credit history, or how you've managed debt in the past.
- Capacity – ability to repay loans.
- Capital – 用来贷款的储蓄 – 买房首付越多 --> better interest rate
- Collateral – 抵押品 – 用于担保
- Conditions – other information e.g. How you plan to use the money

FI risk

Overview: Financial risk

- **Financial Risk** is the possibility of **losing** monetary capital or suffering **negative** financial outcomes. It is the uncertainty associated with any investment, loan, or business decision, representing the potential that actual returns will differ from expected returns. All financial activities involve some degree of risk, and the core principle of finance is that **higher potential returns are generally accompanied by higher levels of risk.**

- **Losing monetary capital** — Refers to money and other financial resources / **Negative financial outcomes**
- Uncertainty – Actual vs expected returns – 投资回报可能偏离原本的估算
- Higher returns = Higher risk

Key financial risk

1.1. Market risk

Market risk: Exposure to the uncertain market value of a portfolio, where the underlying economic factors are such as interest rates, exchanges and equity and commodity prices.

投资组合的市场价值因**经济因素**波动而产生的不确定性，而这种不确定性来自于利率、汇率、股票和商品价格等经济因素的波动。

Exposure to the uncertain market value of a portfolio 投资组合

- Key Question: What if the market price of my asset goes down? --> 体现了市场风险的核心：价格波动带来的潜在损失

Equity Risk 股权风险	股票价格下跌，导致投资亏损
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Interest Rate Risk	利率上升 → Bond 债券价格下跌
Currency Risk	外币投资回本时汇率不利，导致实际收益减少
Commodity Risk	The price of oil/gold/grains changes.

- Bond? – 债券

- **Bond** is a type of loan—but in reverse.
- You **lend money** to a company or government, and they promise to **pay you back later** with interest.
- **债券**是一种融资工具，本质上是你把钱借给发行方（公司或政府），他们承诺在未来某个时间还本付息。

1.2. Credit risk

- **Credit risk:** The risk that a borrower or counterparty will fail to meet their obligations.

Key Question: What if the person/company I lent money to doesn't pay me back?

公司违约	企业未能偿还公司债券，投资者损失本金和利息
个人违约	借款人未能按时支付房贷 mortgage，银行面临坏账风险
衍生品违约	衍生合约中的交易对手未能履行付款义务，造成交易失败

- 一家企业发行了债券向公众或机构借钱，但在约定的时间内**无法支付利息或归还本金**，也就是**违约 (default)**了。

1.3. Liquidity risk 流动性风险

- **Liquidity risk:** The risk of being unable to meet cash flow needs or to buy/sell an asset quickly without causing a significant move in its price.

无法及时满足现金流需求，或资产迅速买卖而引发价格剧烈波动的风险。

What if I need **cash** urgently and can't sell my assets?

What if depositors 存款人 all want their money back at once?

Asset Liquidity	Needing to sell a rare painting or large block of stock quickly; may require a deep discount.	急售稀有画作或大量股票，可能要大幅降价才能成交。
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Funding Liquidity	A bank faces a "bank run" where too many depositors withdraw at once.	银行遭遇“挤兑”，储户集中提款，资金链断裂。
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1.4. Operational risk

- Operational risk: The risk of loss resulting from failed internal processes, people, systems, or from external events. It is not driven by market or credit factors.

因**内部**流程、人员、系统失败或**外部**事件导致的损失风险，不受市场或信用因素驱动。

What if our computer system fails, an employee makes a mistake, or a natural disaster occurs?

– 如果我们的系统崩溃、员工出错或发生自然灾害怎么办？

Data Breach	A data breach exposes customer information. 数据泄露导致客户信息外泄。
Human Error	A key employee makes a costly error.
Natural Disaster	A factory is damaged by a natural disaster.
Internal Fraud	A rogue trader causes unexpected losses. 内部人员违规交易，造成损失。

Measuring of Financial Risk

1. Measuring Market Risk

1.1. Sensitivity Analysis (敏感性分析)

Measures how the **value of a portfolio** changes given a small change in a single underlying market risk factor (e.g., interest rates, stock index).

评估当某个市场因素（如利率、股指）发生微小变动时，投资组合的价值如何变化。

1.2. Value at Risk (VaR) (风险价值)

Estimates the **maximum potential loss** over a specific time period (e.g., 1 day) at a given confidence level (e.g., 95% or 99%).

在特定时间段内（如一天），以一定置信水平（如95%或99%）估算最大可能损失。

1.3. Example Scenarios (情境模拟)

情境模拟是一种风险评估技术，它通过构建假设性的极端市场事件，来测试投资组合或金融系统在这些情况下的表现。

通俗说法：就是“假如真的发生了大事，我的资产会怎么样？”

➤ Example Scenarios

- 2008-style market crash.
- A sudden 5% rise in interest rates.
- A specific geopolitical event. — 如战争爆发、制裁升级、能源供应中断等。

2. Measuring Credit Risk (衡量信用风险)

2.1. Credit Scoring (信用评分)

A statistical model used primarily for retail borrowers (individuals, small businesses) to predict the **probability of default** 违约概率 based on historical data 主要用于个人或小型企业的统计模型，通过历史数据预测违约概率。

2.2. Credit Ratings (信用评级)

Assessments provided by rating agencies (S&P, Moody's, Fitch) **for corporations and governments**. They provide an opinion on the entity's **creditworthiness**.

- 指的是一个实体 (entity) 偿还债务的能力和可信度。

Assessing creditworthiness	评估信用状况
High creditworthiness	高信用等级 → 借款风险低，利率低
Low creditworthiness	低信用等级 → 借款风险高，可能被拒贷或利率高

<https://www.investopedia.com/ask/answers/110614/whats-difference-between-credit-rating-and-credit-score.asp#toc-key-differences>

Credit scores are three-digit numbers that tell lenders whether an individual is likely to be a responsible borrower. Credit ratings, on the other hand, are letter ratings assigned to corporations or governments and are used by investors to determine their riskiness.

2.3. Probability of Default (PD) (违约概率)

The likelihood that a borrower will default over a specific time horizon (e.g., 1 year). 是高级信用风险模型的核心指标。

A core component of advanced models. – 借款人在特定时间内（如一年）发生违约的可能性

3. Measuring Liquidity risk

- Asset Liquidity Risk（资产流动性风险）-> 这是指资产在需要变现时，是否能快速出售而不引发价格剧烈波动。
 - Stress Testing（压力测试）

Simulating scenarios like a credit downgrade, which may trigger collateral calls and large cash outflows, to see if the institution can survive.

假设公司信用变差了（比如被评级机构降级），这可能会导致别人要求你立刻补交更多抵押品，还可能要你马上拿出很多现金。我们通过这种“假设情境”来测试公司能不能撑住，不会破产或资金链断裂。

4. Measuring Operational risk

Basel II（巴塞尔协议II）

这是国际银行监管框架，提出了三种**衡量操作风险**的方法：

- Basel II proposed three alternatives to measure operational risks:
 - Basic Indicator Approach (BIA)
 - Standardized Approach (TSA)
 - Advanced Measurement Approaches (AMA)

Basic Indicator Approach	BIA	基本指标法：用总收入的一定比例估算操作风险。
Standardized Approach	TSA	标准法：按业务线划分，每类业务用不同系数计算风险。
Advanced Measurement Approach	AMA	高级法：使用内部模型、历史数据和场景分析来精算风险。

Basel II（巴塞尔协议 II） 所定义的七类**操作风险事件类型（Operational Risk Event Types）**，用于帮助金融机构识别和分类内部可能发生的非市场性损失



- The Basel II defined event types for each category:
 - ❖ Category 1: Internal Fraud
 - ❖ Category 2: External Fraud
 - ❖ Category 3: Employment Practices and Workplace Safety
 - ❖ Category 4: Clients, Products & Business Practices
 - ❖ Category 5: Damage to Physical Assets
 - ❖ Category 6: Business Disruption & System Failures
 - ❖ Category 7: Execution, Delivery & Process Management

Internal Fraud	内部欺诈	<i>Fraudulent acts by employees, such as embezzlement or insider trading.</i> ✚ <i>Example: A bank employee secretly transfers client funds to a personal account.</i>
External Fraud	外部欺诈	<i>Criminal acts from outside the organization, like cyberattacks or theft.</i> ✚ <i>Example: Hackers breach a bank's system and steal customer data.</i>
Employment Practices & Workplace Safety	雇佣与职场安全	<i>Issues related to employee treatment, discrimination, or unsafe conditions.</i> ✚ <i>Example: A lawsuit over workplace harassment or injury due to poor safety.</i>
Clients, Products & Business Practices	客户、产品与业务行为	<i>Misconduct in delivering products or services, including mis-selling or breach of regulations</i> ✚ <i>Example: Selling unsuitable investment products to elderly clients.</i>
Damage to Physical Assets	实体资产损毁	<i>Losses due to natural disasters, vandalism, or accidents.</i> ✚ <i>Example: A flood damages a data center, disrupting operations.</i>
Business Disruption & System Failures	业务中断与系统故障	<i>Interruptions caused by IT failures, power outages, or telecom breakdowns.</i> ✚ <i>Example: A trading platform crashes during peak hours, halting transactions.</i>

Execution, Delivery & Process Management	执行、交付与流程管理	<i>Errors in transaction processing, data entry, or supply chain mismanagement.</i> ✚ <i>Example: A bank processes a loan with incorrect interest terms due to system input error.</i>
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Disaster Planning

What is disaster?

- **Disaster**
 - Cause great damage, loss of lives and destruction.
 - Occur throughout the world, in every form of natural, **man-made**, and combination of disaster.
 - Unexpected, and tax the ability of governments and other agencies to cope.

1. Loss
2. Natural / man-made / Combination
3. Unexpected
4. 会严重消耗政府和相关机构的应对能力，甚至让他们应接不暇。“tax”并不是“征税”的意思，而是一个动词用法，表示“给……造成沉重负担”或“极度考验”

Emergency Management?

- **Emergency Management**
 - Identification of what is not known;
 - Collection – gathering information related to what is not known;
 - Production – answering management questions;
 - Dissemination – getting the answers to the right people.

1. 识别未知 (Identification of what is not known)
 - **核心意义**：不是先解决问题，而是先承认“我们不知道什么”。
 - 比如灾情范围、受影响人群、资源缺口。
2. 收集信息 (Collection)
 - **核心意义**：围绕未知展开调查，收集相关数据、文献、反馈等。
 - 例如通过无人机或社交媒体收集灾区图像和求助信息，补足官方数据。

3. 生成答案 (Production)

- **核心意义：**将收集到的信息转化为可操作的结论或建议。
- 将收集的信息转化为决策支持——如制定疏散路线、资源分配方案、医疗优先级。

4. 信息传达 (Dissemination)

- **核心意义：**确保正确的人在正确的时间收到正确的信息。
- 通过广播、短信、社交平台等多渠道发布信息，避免恐慌与误解